

Review

Immobilization of Cadmium, Lead, and Copper in Soil Using Bacteria: A Literature Review

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Abstract

The heavy metal contamination of soils is a global environmental challenge threatening water quality, food safety, and human health. Using a systematic literature review approach, this study aimed to assess the potential of bacterial strains to immobilize cadmium (Cd^{2+}), lead (Pb^{2+}), and copper (Cu^{2+}) in contaminated soils. A total of 45 articles were analyzed, focusing on studies that reported heavy metal concentrations before and after bacterial treatment. The analysis revealed that bacterial genera such as *Bacillus*, *Pseudomonas*, and *Enterobacter* were most commonly used for the immobilization of these metals. Immobilization efficiencies ranged from 25% to over 98%, with higher efficiencies generally observed when microbial consortia or amendments (e.g., phosphate compounds and biochar) were applied. The main immobilization mechanisms included biosorption, bioprecipitation (such as carbonate-induced precipitation), bioaccumulation, and biomineralization, which convert mobile metal ions into more stable, less bioavailable forms. These findings highlight the promising role of microbial-assisted immobilization in mitigating heavy metal pollution and reducing ecological risks. Further laboratory and field studies are needed to optimize the use of these microbial strains under site-specific conditions to ensure effective and sustainable soil remediation practices.

Keywords: heavy metals; bacterial immobilization; cadmium; lead; copper; biosorption; soil remediation; environmental sustainability



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1. Introduction

More than 20 million hectares of land worldwide are contaminated with heavy metals [1]. The remediation of contaminated soil requires substantial investments, and complete soil recovery may take between 200 and 1000 years [2]. Heavy metals are biologically non-degradable, making them difficult to remove from soil and leading to their accumulation in living organisms [3]. Globally, these metals are of serious concern due to their persistence and stability in the environment [4]. Heavy metal contamination has increased due to various industrial activities, growing amounts of waste, and modernized agricultural practices [5]. The scientific literature reports that cadmium (Cd), lead (Pb), and copper (Cu) are among the primary toxic metals that contaminate soil [6,7]. The concentrations of these metals in soils vary: lead, from 1.5 to 663 mg kg⁻¹; cadmium, from 3.6 to 300 mg kg⁻¹; and copper, from 1.8 to 32 mg kg⁻¹ [8].

Soil pollution poses a significant risk to water quality and food safety, as heavy metals can contaminate crops and enter the human body through the food chain [9]. Long-term exposure to heavy metals negatively affects human health and can lead to various cancers,

nervous system disorders, insomnia, and other illnesses [9]. The toxic effects of these metals on living organisms even occur at low concentrations, underscoring the need for their removal to immobilize them from the environment [6]. The use of microorganisms to mitigate heavy metal contamination is a promising strategy. However, it remains essential to clarify what bacteria alter their chemical forms to reduce toxicity and mobility.

Biological remediation methods can be effective and provide long-term results, as they use living systems that can adapt to different soil conditions [10]. The use of microorganisms to remove heavy metals is a promising strategy to address the consequences of soil contamination caused by human activities [3]. The concentration of heavy metals in soil significantly influences microorganism activity, survival, and remediation potential, often inhibiting key microbial processes. Immobilization is an effective method for the large-scale restoration of heavy-metal-contaminated soils [11]. The effectiveness of microorganisms can depend on the soil type and its chemical composition and environmental conditions, including pH, temperature, and moisture. Depending on the microorganism, it may be effective at immobilizing certain heavy metals but not others [12]. Traditional remediation methods are often costly and environmentally invasive. Therefore, there is growing interest in eco-friendly and cost-effective alternatives such as microbial-assisted immobilization. Understanding how microorganisms behave under different contamination levels and environmental conditions is crucial for optimizing their application in real-world scenarios [5].

Several processes performed by microorganisms contribute to heavy metal immobilization. Biodegradation involves the immobilization of heavy metals by microorganisms. Biosorption occurs when heavy metals are adsorbed onto the cell walls of microorganisms. Biological precipitation refers to the formation of insoluble metal compounds. Biotransformation involves the chemical transformation of metal compounds through reduction or oxidation reactions. Bioaccumulation occurs when microorganisms accumulate heavy metals in their cells, making this applicable in bioremediation processes. One of the main mechanisms by which microorganisms remove metals is bioremediation, a process where environmental contaminants are broken down or transformed into less hazardous compounds. Bioaugmentation, a bioremediation strategy, involves introducing effective microorganisms into the environment to facilitate the breakdown or transformation of pollutants [13].

Bacillus species are gram-positive, spore-forming, rod-shaped bacteria that are either aerobic or facultative anaerobes. *Bacillus* species are capable of forming spores under extreme conditions. Due to their specific structure, the spores can resist high temperatures, drought, moisture, radiation, and the effects of heavy metals [14]. *Pseudomonas* species are gram-negative, rod-shaped, motile bacteria that are strictly aerobic but metabolically versatile. They are known for their high adaptability to various environments, including contaminated soils, and for producing siderophores and biosurfactants that contribute to heavy metal chelation and immobilization. *Pseudomonas* spp. exhibit resistance to toxic metals and are frequently used in bioremediation due to their efficient metal transformation capabilities [15]. *Enterobacter* species are gram-negative, facultative anaerobic, rod-shaped bacteria commonly found in soil and water. They can tolerate heavy metal stress and contribute to metal immobilization through biosorption, precipitation, and enzyme-mediated redox transformations. *Enterobacter* spp. are also capable of forming biofilms, which enhance their survival and remediation potential in contaminated environments [16]. Immobilization through ion exchange reactions alters the adsorption, complexation, and precipitation of heavy metals from active to stable phases, thereby reducing their bioavailability in soil [17]. However, it is essential to identify, based on the scientific literature, which microorganisms can effectively immobilize all three of the aforementioned heavy metals.

2. Materials and Methods

This study applied a scientific literature review methodology to examine the role of soil microorganisms in the immobilization of heavy metals, specifically cadmium (Cd^{2+}), lead (Pb^{2+}), and copper (Cu^{2+}). The aim was to evaluate how many recent scientific studies on this topic could be identified using an open-access tool, and this approach ensured the inclusion of freely accessible publications. The literature review covered 45 scientific articles published between 2006 to 2025. The inclusion criteria for the analysis were the following:

1. Scientific articles investigating the use of bacteria for the immobilization of cadmium, lead, and copper in contaminated soils.
2. Articles that presented data on the concentrations of these heavy metals before and after treatment with bacteria.
3. Studies describing the bacterial strains or consortia used for heavy metal immobilization.

Articles not meeting the criteria or not directly related to heavy metal immobilization by bacteria in soil were excluded from the analysis. The following data were extracted from the included studies: concentrations of cadmium, lead, and copper before and after bacterial treatment; bacterial strains used for immobilization; the reported percentage reduction in heavy metal concentrations; and conditions and duration of the experiments, where available.

This review aims to systematically summarize the efficiency of bacterial strains in immobilizing cadmium, lead, and copper in soil, while also evaluating the limitations posed by varying environmental conditions and discussing practical applications. This integrated approach provides new insights into microbial-assisted remediation beyond laboratory conditions.

Using the systematic literature review method, a search and an analysis of literature sources were conducted to systematize and describe the possibilities for immobilizing heavy metals in soil using bacteria. In this work, we searched for peer-reviewed, open-access, full-text scientific articles in English that presented studies on the immobilization of heavy metals in soil. The systematic literature review included full-text articles found in two scientific databases, PubMed and ScienceDirect, that investigated soil contamination and the potential for heavy metal immobilization using bacteria.

The literature search was performed from 10 March 2025, to 24 March 2025. The main English keywords used in the search in these scientific databases were heavy metals, immobilization, bacteria, and soil. The search initially used the keyword combination [“heavy metals in soil” and “heavy metal immobilization from soil”]. Later, the search was further refined by adding the keyword “using bacteria” to the combination. Using these combinations in the two scientific databases identified 715 publications.

Initially, the titles of the articles were screened, followed by the abstracts, and finally the full texts. During the title screening phase, it was assessed whether the publications addressed the potential of bacteria for immobilizing heavy metals in soils. While various microbial-based remediation strategies exist, this review focuses on immobilization because it offers stable, long-term, continuous bacterial activity or reapplication. Publications that did not match the topic were excluded from further consideration (Figure 1).

The collected data are summarized and presented in tables, displaying the concentrations of heavy metals in soil before and after immobilization, the bacterial strains used for immobilization, and the immobilization efficiency as a percentage reduction in heavy metal concentrations. As this research is based solely on a literature analysis and did not involve new laboratory experiments, animal or human subjects, or interventionary studies, no ethical approval was required. No new data sets were generated, and all information used in this work is publicly available from the materials or information reported in the review.

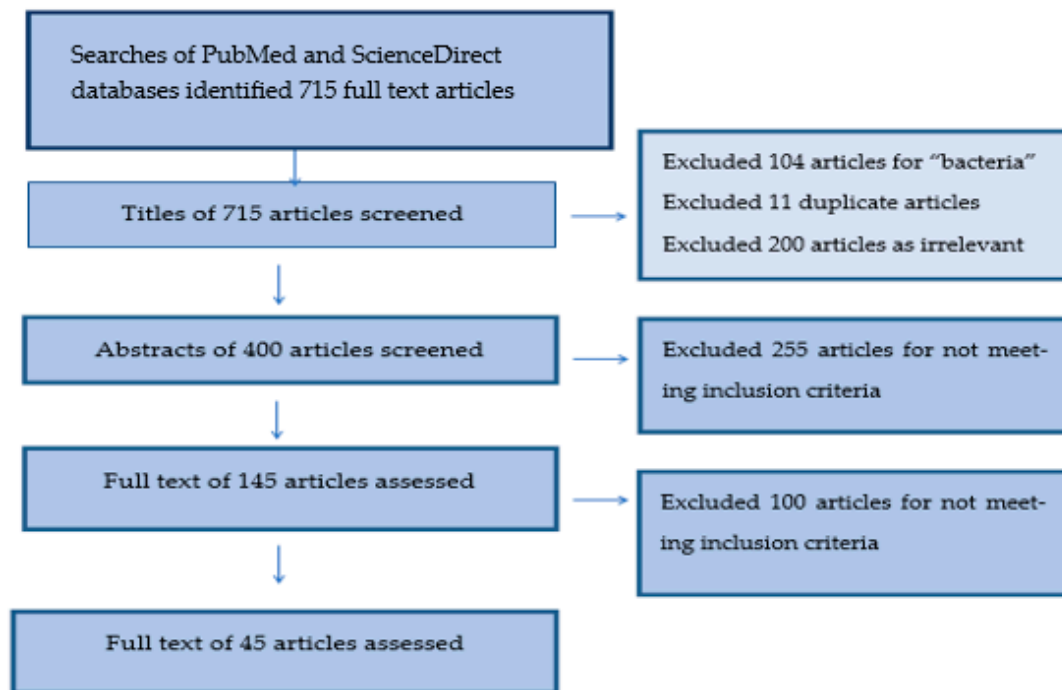


Figure 1. Flowchart of the screening process for systematic literature review.

However, it is important to note that the results were obtained under differing environmental conditions, including variations in pH, temperature, and soil composition. Therefore, comparisons between bacterial strains should be interpreted cautiously, as efficiency may not be solely attributable to the strain itself but also to the experimental setup.

Nonetheless, these efficiencies were reported under varied experimental conditions. As such, while comparisons provide insights into potential effectiveness, they may not fully reflect performance under standardized conditions. Additional research under controlled, uniform setups is required for direct comparative assessment.

Different environmental conditions and experimental designs were used in these studies, which limits the comparability of bacterial efficiency data across the reviewed literature. Although trends can be identified, direct efficiency comparisons should be made with caution unless environmental parameters are standardized.

In most of the reviewed studies, heavy metal concentrations were determined using standard instrumental techniques, primarily inductively coupled plasma optical emission spectrometry (ICP-OES) and, in some cases, ICP-MS.

This work aims to facilitate further laboratory studies related to the immobilization of heavy metals in soil using bacteria.

3. Results

With the advancement of modern microbial technologies, significant progress has been made in adapting microorganisms to tolerate heavy metals and in isolating specific genes that can enhance the specificity and efficiency of heavy metal removal [18]. Microorganisms can adapt to a variety of environmental conditions, making them widely applicable for the reduction in heavy metal contamination in soils. These characteristics include a broad source base, rapid growth, high tolerance, and high efficiency [19]. For this reason, scientific articles on this topic were analyzed to identify which bacteria are capable of immobilizing cadmium, lead, and copper, as well as to determine which bacteria are most commonly used by researchers and their respective effectiveness.

Figure 2 (Table S1) presents studies on the application of various microorganisms, assessing their ability to immobilize Cd^{2+} in contaminated soils or other environments. It includes details such as sampling locations, contamination levels, achieved immobilization efficiencies, and the bioremediation methods applied.

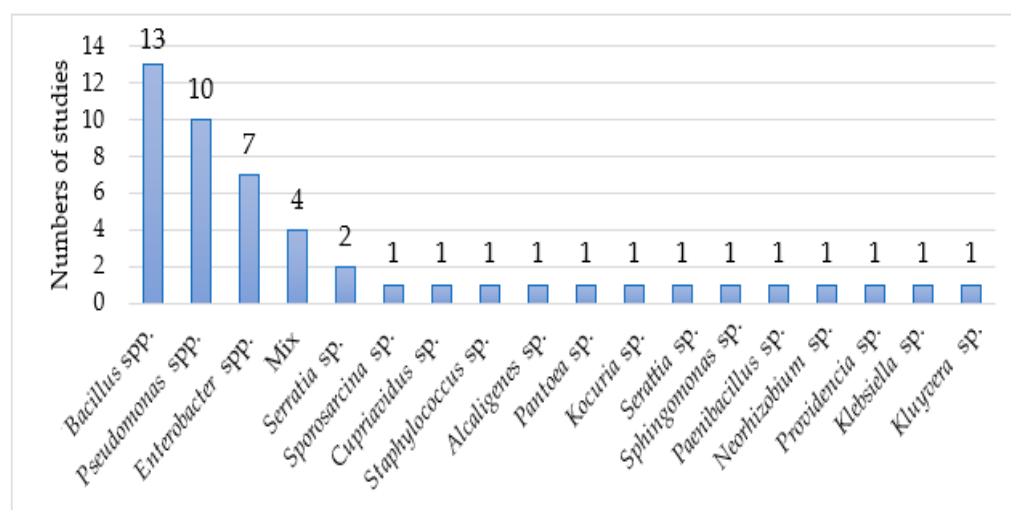


Figure 2. Number of studies using specific bacterial genera (e.g., *Bacillus*, *Pseudomonas*, *Enterobacter*, etc.) investigating cadmium immobilization in contaminated soils. The figure illustrates the number of peer-reviewed articles that employed each genus in experimental bioremediation settings.

Figure 2 illustrates the frequency of microbial genera used in various scientific studies that analyzed their effectiveness in reducing cadmium mobility or bioavailability in soil. The most frequently used genus was *Bacillus* spp. (13 studies), known for its high resistance to heavy metal stress and its ability to form spores, which enables survival under unfavorable conditions. *Pseudomonas* spp. ranked second (10 studies), with representatives well known for their metabolic versatility and ability to synthesize biosurfactants. *Enterobacter* spp. was mentioned in seven studies, while mixed microbial consortia were used in four. The remaining genera, such as *Serratia*, *Cupriavidus*, *Providencia*, and others, were used only in individual studies (once each), indicating their less frequent application or lower research prevalence in this field.

Figure 3 presents cadmium concentrations in various soil types collected from different locations.

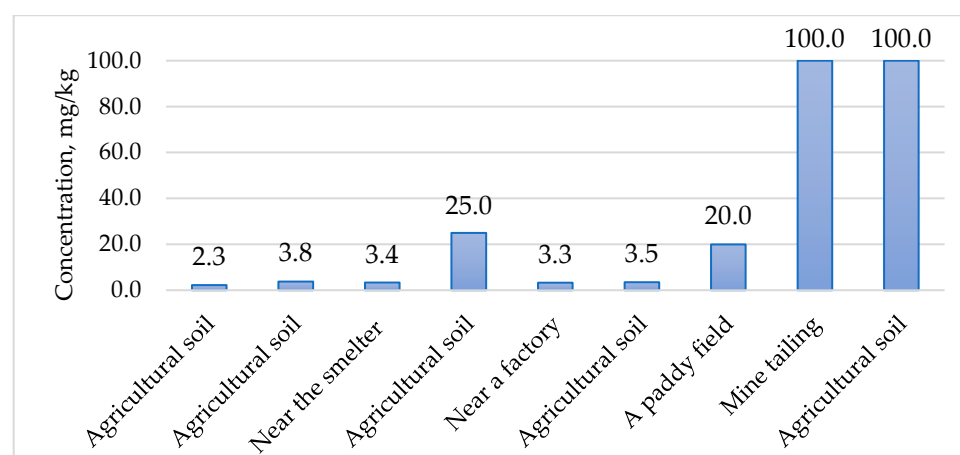


Figure 3. Cadmium concentrations in contaminated soils investigated in scientific studies, mg/kg.

In most agricultural soils, cadmium concentrations ranged from 2.3 to 3.8 mg/kg. Similar levels were found in soils near a smelter and a factory (3.4 and 3.3 mg/kg, respectively). Significantly higher cadmium concentrations were recorded near one agricultural area (25.0 mg/kg) and in a paddy field (20.0 mg/kg), which may indicate a local source of pollution or the intensive use of agrochemicals. The highest cadmium concentrations were found in mine tailings and in one of the agricultural soil samples—reaching as high as 100.0 mg/kg. These values significantly exceed typical background levels, indicating severe contamination and posing a potential risk to human health and ecosystems.

Figure 4 illustrates the cadmium immobilization efficiency of different *Bacillus* species, expressed as percentages.

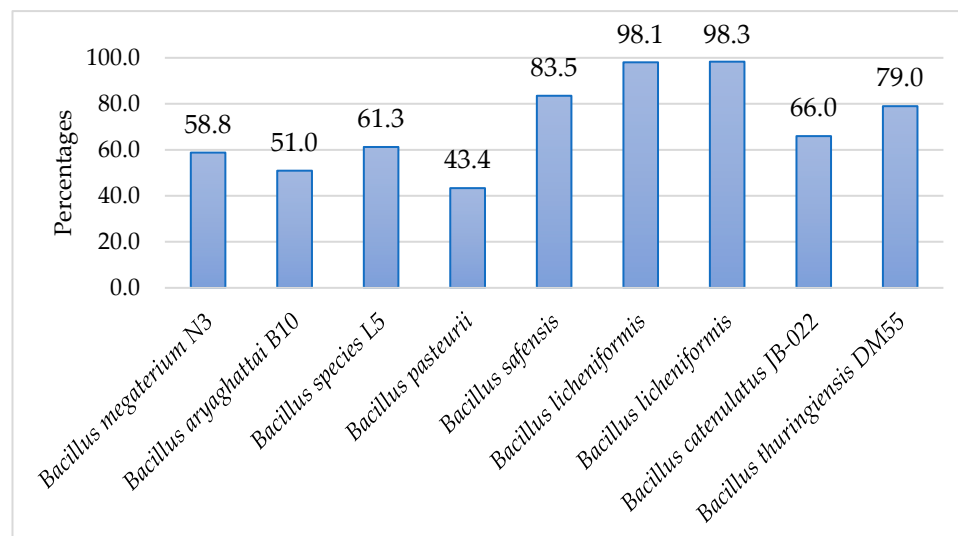


Figure 4. The efficiency of cadmium immobilization in soil using *Bacillus* species, %.

The highest efficiency was observed for *Bacillus licheniformis* (98.1% and 98.3%) and *Bacillus safensis* (83.5%), which significantly outperformed the other tested species. This suggests that these microorganisms have exceptional potential for application in the bioremediation of contaminated soils. In contrast, *Bacillus pasteurii* (43.4%) and *Bacillus aryabhattai* B10 (51.0%) demonstrated lower immobilization efficiency. Other species, such as *Bacillus megaterium*, *Bacillus cereus* JB-022, and *Bacillus thuringiensis* DM55, exhibited moderate efficiency, ranging from 58.8% to 79.0%.

The duration of cadmium immobilization by *Bacillus* species varied between studies, ranging from a few days to several weeks. However, long-term stability depends on environmental conditions and whether bacteria remained viable or entered spore form.

Figure 5 presents the cadmium immobilization efficiency, expressed as percentages, using different *Pseudomonas* species.

The highest efficiency was observed for *Pseudomonas aeruginosa* (87.0% and 85.0%) and *Pseudomonas fluorescens* (83.0%), suggesting that these microorganisms possess strong potential for cadmium bioremediation applications. In contrast, *Pseudomonas koreensis* (48.5%), *Pseudomonas plecoglossicida* (G1), and *Pseudomonas taiwanensis* (P4) (both 51.0%) exhibited lower immobilization efficiency, which may be attributed to reduced resistance to heavy metal stress or lower biosorption capacity.

The highest efficiency was observed for *Bacillus licheniformis*; however, it is important to note that the experimental conditions (e.g. pH, soil type, initial Cd concentration) varied across studies. Thus, direct comparisons should be interpreted with caution. The same applies to *Pseudomonas* species evaluations.

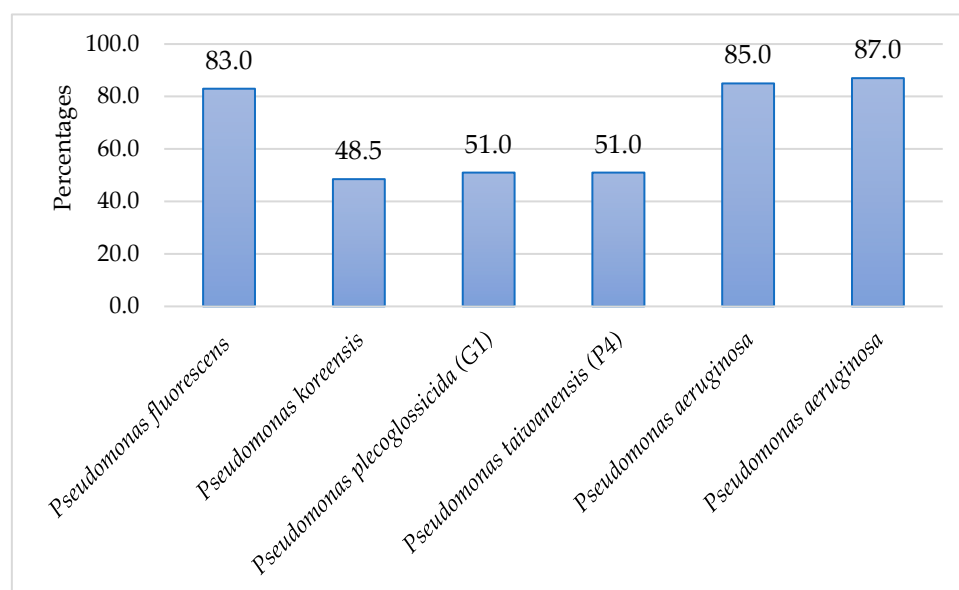


Figure 5. The efficiency of cadmium immobilization in soil using *Pseudomonas* species, %.

Figure 6 illustrates the cadmium immobilization efficiency of bacteria belonging to the *Enterobacter* genus, with values expressed as percentages.

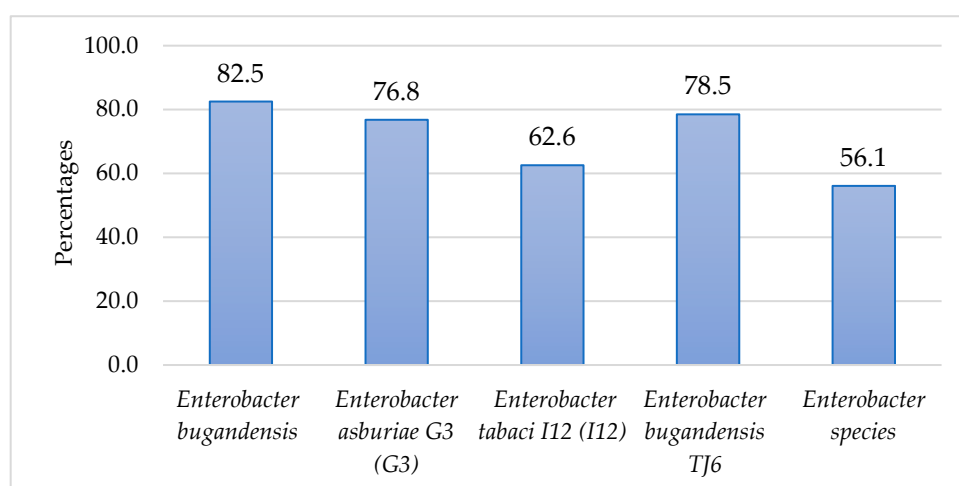


Figure 6. The efficiency of cadmium immobilization in soil using *Enterobacter* species, %.

The highest efficiency was observed for *Enterobacter bugandensis* (82.5%) and its strain TJ6 (78.5%), as well as *Enterobacter asburiae* G3 (76.8%), indicating strong potential for the application of this genus in bioremediation. Slightly lower efficiency was recorded for *Enterobacter tabaci* I12 (62.6%), while the lowest was observed for an unspecified *Enterobacter* species (56.1%).

The results of the literature analysis demonstrate that various microorganisms, such as *Bacillus*, *Pseudomonas*, *Enterobacter*, and *Providencia* species, have been successfully applied for cadmium immobilization in different contaminated sites, including mining wastes, agricultural soils, and landfills. The immobilization efficiency ranges from approximately 25% to as high as 98%, depending on the microbial strains used, environmental conditions, and the presence of additional amendments (e.g., biochar, phosphate compounds, and carbonates). These findings indicate that microbially driven bioremediation is a promising approach that can be tailored for application in various cadmium-contaminated environments.

Figure 7 (Table S2) presents studies on the application of various microorganisms, assessing their ability to immobilize Pb^{2+} in contaminated soils or other environments.

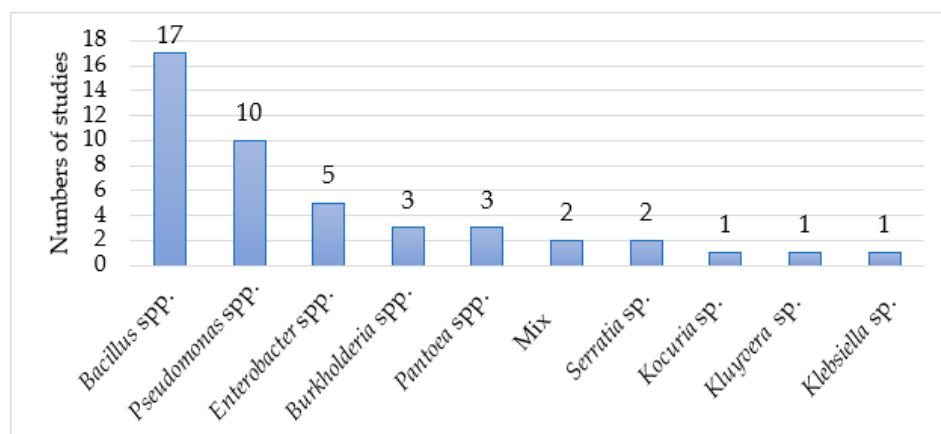


Figure 7. Number of studies using specific bacterial genera on lead immobilization. This figure summarizes how often each genus (e.g., *Enterobacter*, *Burkholderia*, *Pantoea*) appeared in published articles investigating Pb^{2+} immobilization, reflecting the extent of research focus on each.

Enterobacter spp. was mentioned in five studies, while *Burkholderia* spp. and *Pantoea* spp. were each reported in three studies, reflecting a moderate level of research intensity. The use of mixed microbial cultures (Mix), as well as *Serratia* spp., appeared in two studies. Less commonly used genera, such as *Kocuria* sp., *Kluyvera* sp., and *Klebsiella* sp., were each reported only once, which may indicate their lower application prevalence or insufficient investigation in the context of lead immobilization.

Figure 8 presents the lead concentrations in soil samples collected from various locations, including agricultural areas and sites near factories.

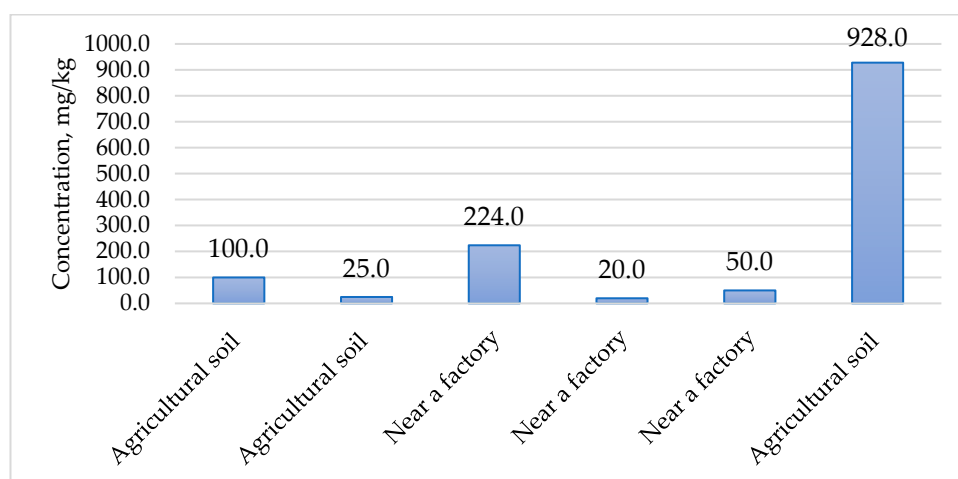


Figure 8. Lead concentrations in contaminated soils investigated in scientific studies, mg/kg.

In most agricultural soil samples, lead concentrations ranged from 25.0 to 100.0 mg/kg, which may be associated with atmospheric deposition, pesticide use, or former industrial activities. A concentration as high as 224.0 mg/kg was detected in soil near a factory, indicating the significant accumulation of contaminants due to industrial processes. An exceptionally high concentration—928.0 mg/kg—was found in one of the agricultural soil samples, raising serious concerns about potential heavy metal contamination.

Given that environmental parameters such as pH, temperature, and soil composition differed among studies, direct comparison of lead immobilization efficiency between bacterial species must account for these variables.

Figure 9 illustrates the lead immobilization efficiency, expressed as percentages, using various *Bacillus* species.

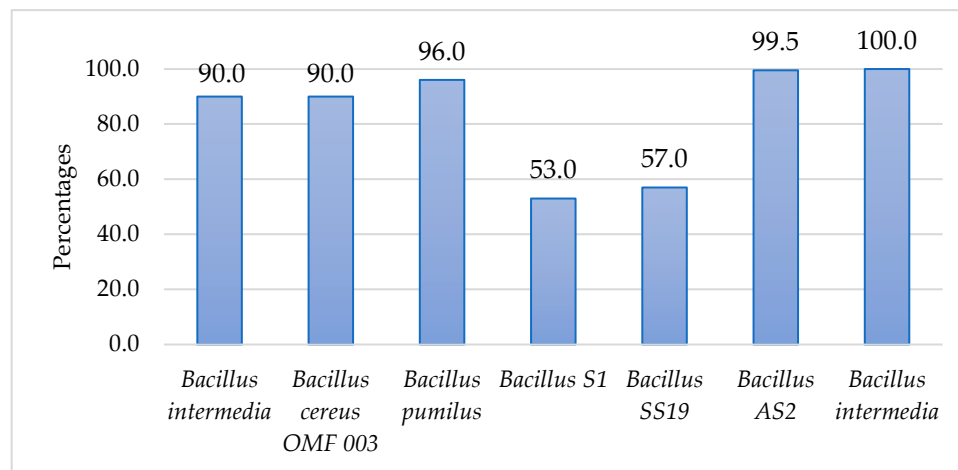


Figure 9. The efficiency of lead immobilization in soil using *Bacillus* species, %.

The highest efficiencies were observed for *Bacillus intermedia* (100.0% and 90.0%), *Bacillus AS2* (99.5%), and *Bacillus pumilus* (96.0%), indicating strong potential for lead retention in soil. These results suggest that certain *Bacillus* strains may be particularly effective in bioremediation processes related to lead contamination. In contrast, *Bacillus S1* (53.0%) and *Bacillus SS19* (57.0%) exhibited lower immobilization efficiency, which may be related to reduced biosorption capacity or greater sensitivity to heavy metal stress.

Figure 10 illustrates the lead immobilization efficiency, expressed as percentages, using various *Pseudomonas* species.

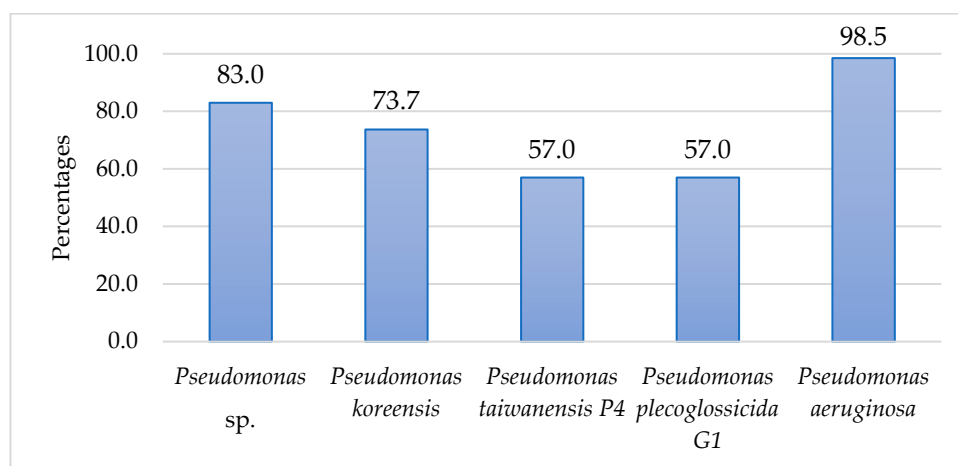


Figure 10. The efficiency of lead immobilization in soil using *Pseudomonas* species, %.

The highest efficiency was observed for *Pseudomonas aeruginosa* (98.5%), indicating the exceptional potential of this microorganism in bioremediation processes. High efficiency was also demonstrated by *Pseudomonas* sp. (83.0%) and *Pseudomonas koreensis* (73.7%). In contrast, *Pseudomonas taiwanensis* P4 and *Pseudomonas plecoglossicida* G1 showed significantly lower efficiency (both 57.0%), which may be related to lower resistance to lead stress or weaker biosorption mechanisms.

The results of the literature analysis demonstrate that *Bacillus* and *Pseudomonas* genera were the most frequently applied microorganisms for lead (Pb) immobilization, achieving exceptionally high efficiencies in some cases (>90%). Both individual microbial strains and microbial consortia were employed, with the latter enhancing the immobilization efficiency. In some instances, additional materials or strategies (e.g., phosphate compounds or carbonate precipitation) were utilized, which significantly improved the immobilization performance. Various experimental conditions are indicated (e.g., lead concentration ranges, soil types), allowing for the comparative evaluation of the bioremediation efficiency across different environmental settings.

Several studies included additional amendments that may enhance immobilization. Therefore, comparing bacterial efficiency across studies must consider whether such additives were present.

Figure 11 (Table S3) presents comprehensive data on the use of various microorganisms for copper (Cu^{2+}) immobilization in contaminated soils and other environments.

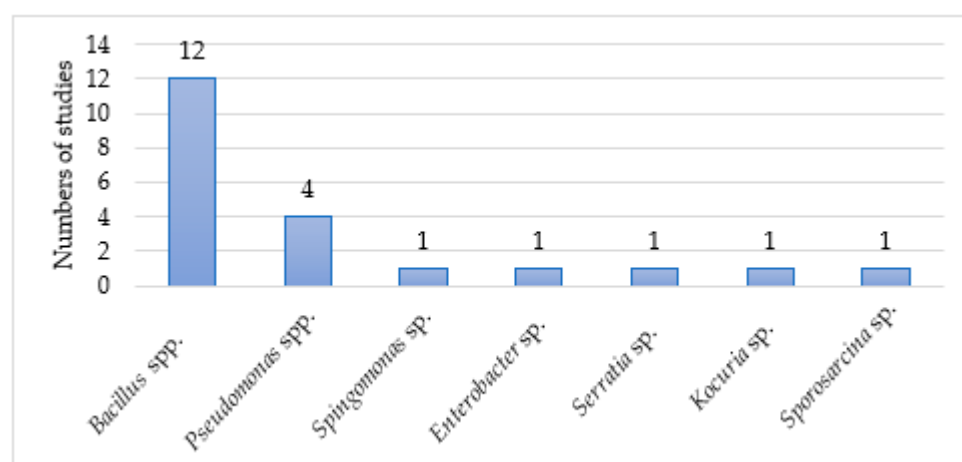


Figure 11. Number of studies using specific bacterial genera on copper immobilization. This figure summarizes how often each genus appeared in published articles investigating Cu^{2+} immobilization, reflecting the extent of research focus on each.

A range of bacterial species, including *Bacillus*, *Pseudomonas*, *Spingomonas*, *Sporosarcina*, *Serratia*, *Kocuria*, and *Enterobacter*, were examined for their potential to immobilize copper through different mechanisms such as biosorption, bioprecipitation, and bioaccumulation.

Tables S1–S3 include detailed information on experimental conditions such as contamination levels, bacterial strains, and immobilization mechanisms. These differences were taken into account during the analysis but the variability remains a limiting factor in direct efficiency comparisons.

Figure 12 illustrates the copper immobilization efficiency, expressed as percentages, using various *Bacillus* species.

The reported copper immobilization efficiencies varied substantially, from as low as 27.3% to as high as 91.8% depending on the microbial strain, contamination level, and remediation conditions. Notably, strains such as *Bacillus thuringiensis* OSM29 and *Bacillus subtilis* D215 demonstrated remarkable immobilization efficiencies (>90% and 67%, respectively), highlighting their potential for practical bioremediation applications.

Experimental conditions varied widely, with soil contamination levels ranging from 5 mg/L to 434 mg/kg of copper, and in some cases, no specific contamination level was reported. Additionally, the role of microbial consortia was emphasized in several studies, where enhanced copper immobilization was observed when multiple strains were used in synergy.

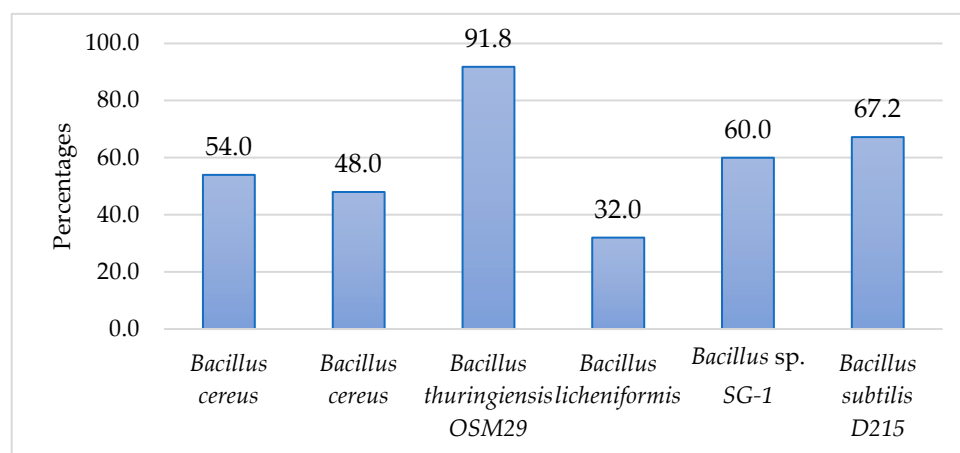


Figure 12. The efficiency of copper immobilization in soil using *Bacillus* species, %.

Table S3 also demonstrates the frequent use of bioprecipitation (e.g., via microbially induced carbonate precipitation, MICP), which in some cases resulted in removal efficiencies above 75%. Other studies leveraged the inherent stress tolerance of certain bacterial strains, such as *Kocuria rhizophila*, to maintain copper immobilization performance in challenging environments.

Moreover, some entries highlight not only copper immobilization but also the accompanying removal of other heavy metals, reflecting the complex and multifaceted nature of real-world contaminated sites. The studies cited in the table provide valuable insights into the adaptability and effectiveness of microbial communities in different environmental contexts.

The addition of amendments like biochar enhances immobilization efficiency by increasing soil pH, surface area, and nutrient availability, which in turn improves microbial habitat and stimulates native microbial activity. These effects create more favorable conditions for microbial metal binding and transformation.

Figure 13 presents a Venn diagram illustrating the ability of microorganisms to immobilize three major heavy metals: cadmium, lead, and copper.

Figure 13 demonstrates how various bacterial species specialize in or share capabilities for immobilizing the heavy metals cadmium, lead, and copper. Each circle represents the immobilization domain of a single metal, while the overlapping regions indicate bacterial species capable of acting on multiple metals. In the central region, universal bacteria that can immobilize all three metals simultaneously are shown, highlighting their potential versatility and high efficacy in bioremediation processes. *Bacillus* sp. is most commonly used to immobilize all three metals.

In conclusion, the data underscore the versatility of microbial remediation strategies for cadmium, lead, and copper immobilization. These findings support the application of tailored microbial consortia or specific strains to address copper contamination in soils, with implications for sustainable and effective bioremediation practices.

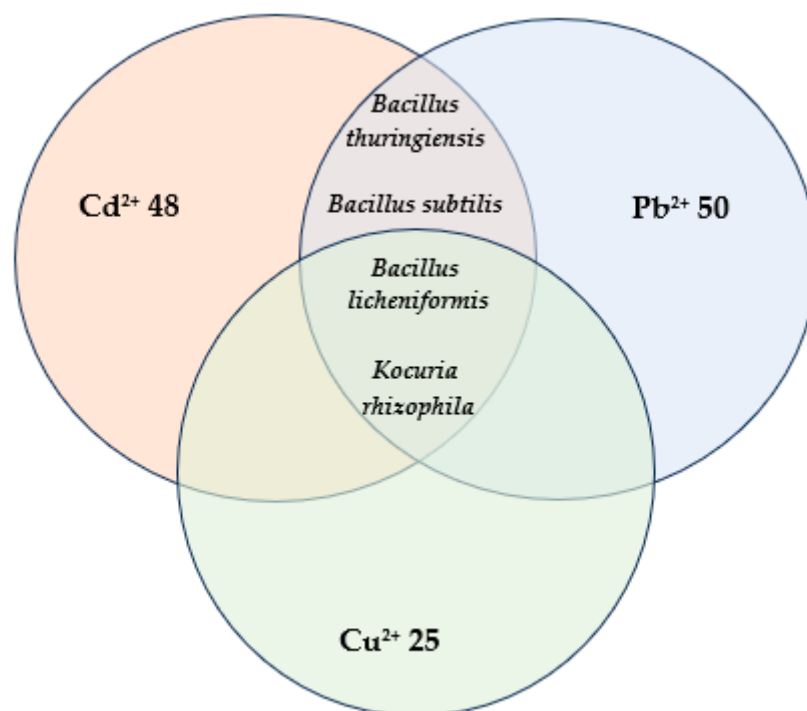


Figure 13. Venn diagram showing bacterial genera capable of immobilizing cadmium, lead, and copper. The overlap indicates genera that can immobilize multiple heavy metals, such as *Bacillus* spp., which is effective against all three. This highlights their versatility in multi-metal bioremediation applications. The efficiency of these bacteria may still vary depending on the metal and environmental conditions.

4. Discussion

The data presented in Tables S1–S3 illustrate the considerable potential of various bacterial strains for immobilizing cadmium (Cd), lead (Pb), and copper (Cu) in contaminated soils and other environmental matrices. Microbial genera such as *Bacillus*, *Pseudomonas*, *Enterobacter*, *Serratia*, and *Providencia* were frequently studied, with some strains demonstrating exceptional immobilization efficiencies (>90%) under optimal conditions. Despite their high efficiency in laboratory settings, the introduced strains often face reduced survival and competitiveness in natural soils due to competition with native microbial communities. Therefore, site-specific selection, adaptation, and even pre-acclimation of strains are necessary for successful field application.

To reduce heavy metal pollution, specific microorganisms that can adsorb, precipitate, or biotransform heavy metals (e.g., *Pseudomonas bacteria*) may be introduced into contaminated soils. The use of microorganisms in remediation can improve the biochemical conditions, create synergistic effects, and ensure the uniform distribution of remediation activity across the soil. When introduced synergistically with indigenous soil microorganisms, these added microbes can enhance overall remediation efficiency. Their addition ensures that the entire treatment process is uniformly distributed throughout the contaminated site [14].

Studies have shown that bacteria can bind to heavy metal ions on their cell walls, thereby preventing these metals from being taken up by plants. Consequently, this reduces the toxic impact of heavy metals on plants, slows down premature leaf senescence, and promotes increased leaf area and pigment content, thereby enhancing photosynthesis [3]. Microbial applications represent a form of treating contaminated environments by modifying the environmental conditions to stimulate microbial growth and promote the removal

of specific pollutants. Microorganisms can reduce the bioavailability of heavy metals by converting them into inactive forms, thereby preventing their uptake by plants [2].

Scientific authors discussed three metal-resistant bacteria: *Bacillus subtilis*, *Paenibacillus jamilae*, and *Pseudomonas aeruginosa*. Among them, *Pseudomonas aeruginosa* was the most effective in mitigating Pb^{2+} toxicity, while *Paenibacillus jamilae* was the most efficient bioremediator for Cd^{2+} or combined $Pb^{2+} + Cd^{2+}$ contamination. This effectiveness was attributed to a noticeable decrease in the uptake of Cd^{2+} and/or Pb^{2+} through plant roots and the reduced translocation of these metals to above-ground plant tissues, leading to improved plant performance, growth, and biomass under heavy metal toxicity [20]. For cadmium, the microbial immobilization efficiencies ranged from approximately 25% to 98%, with *Bacillus* and *Pseudomonas* genera often showing the highest performance. For bacteria-induced carbonate precipitation to be effective, a near-neutral to alkaline pH is typically required to promote carbonate formation and heavy metal precipitation. Many studies also noted that microbial consortia outperformed individual strains, indicating synergistic interactions among different species, while the presence of amendments, such as phosphate compounds and biochar, further enhanced immobilization efficiency. The mechanisms involved in cadmium immobilization included biosorption, bioaccumulation, bioprecipitation, and biomineralization. Lead immobilization followed a similar pattern. Several bacterial strains, including *Bacillus cereus*, *Enterobacter asburiae*, and *Pseudomonas aeruginosa*, achieved immobilization efficiencies exceeding 90%. The formation of insoluble metal phosphates and carbonate-induced precipitation (MICP) were key mechanisms in reducing Pb bioavailability and uptake by plants. These findings underscore the capacity of bacteria not only to adsorb Pb on cell walls but also to precipitate it as stable mineral forms.

Another prominent group is *Bacillus* species, which are gram-positive, spore-forming, rod-shaped aerobes or facultative anaerobes [21]. *Bacillus* species can form spores under extreme conditions. Due to their specific structure, these spores are resistant to high temperatures, drought, humidity, radiation, and the toxic effects of heavy metals such as lead, copper, and cadmium. Microbial technologies can thus be a viable alternative to conventional heavy metal removal methods, particularly for the preservation of agricultural soils [20].

The primary mechanisms of Pb removal include biosorption, bioaccumulation, biological precipitation, and biomineralization, which transform toxic lead ions into insoluble forms to reduce ecological toxicity. These bacteria have demonstrated success in immobilizing heavy metals such as Cu, Cr, and Pb in soil and mining wastes, achieving removal rates exceeding 50% [22]. The copper immobilization data revealed a broad range of efficiencies, from 27.3% to 91.8%, depending on bacterial strain and the experimental conditions. *Bacillus thuringiensis* OSM29 and *Bacillus subtilis* D215 showed particularly high removal efficiencies (>90%). Microbial mechanisms such as biosorption and efflux systems, supported by specific cellular adaptations (e.g., stress tolerance and biofilm formation), were prominent in copper immobilization. The data also suggest that MICP and other mineralization processes play a crucial role in reducing copper mobility in soils.

Various bacterial species, such as *Bacillus*, *Pseudomonas*, *Enterobacter*, *Flavobacterium*, *Geobacter*, and *Micrococcus*, have been studied for their ability to participate in biosorption processes for removing heavy metal ions from contaminated environments. These studies analyzed bacterial cell surface to volume ratios and other specific features that strengthen the interactions between bacterial functional groups and heavy metal ions [23]. *Bacillus* species are considered among the most effective bacteria for removing lead, cadmium, and copper [24]. Bacterial cell surfaces exhibit various features that prevent metal ions from entering the cell by adsorbing them onto the cell surface and acting as barriers [25]. The biofilms produced by microorganisms are composed of extracellular polymers that

can accumulate metal ions, thus protecting the cells within them from toxic effects [26]. Microbial resistance to metal ions is also supported by enzymes that biologically transform or chemically modify metal ions from highly toxic forms into less toxic states. Metal ion toxicity can be effectively mitigated by altering their redox states through reduction or oxidation reactions [27]. This process is supported by detoxification enzymes whose activity is regulated by microbial resistance genes.

The microbial cell membrane contains various proteins and metabolic products capable of forming complex structures with metal ions. Extracellular sequestration can be defined as a process by which organisms (fungi and bacteria) remove heavy metals by binding them to external cell components [28]. Iron- and sulfur-reducing bacteria such as *Desulfuromonas* and *Geobacter* species can reduce dangerous metals to less hazardous or non-toxic forms. In intracellular sequestration, metal ions are complexed with various compounds in the cytoplasm. The interaction of metal ions with surface ligands and their gradual transformation into the cell can result in significant metal accumulation within microbial cells. This ability of bacterial cells to sequester metals intracellularly is utilized in various applications, especially in waste treatment processes [3].

Future research should focus on strategies to enhance the remediation efficiency of microorganisms, including the development of genetically modified strains, the optimization of consortia for synergistic action, and the application of organic additives that stimulate microbial metabolism. These approaches may improve the persistence and functionality of microbial inoculants in contaminated soils.

Overall, the data support the hypothesis that microbial-assisted immobilization can significantly reduce the bioavailability and ecological risk of heavy metals in contaminated soils. The variability in efficiencies highlights the importance of considering site-specific conditions, microbial adaptability, and potential synergistic effects when designing bioremediation strategies.

5. Conclusions

This literature-based analysis confirms the promising potential of microbial-based bioremediation for the immobilization of cadmium, lead, and copper in contaminated soils. Among the analyzed bacterial strains, *Bacillus* spp. and *Pseudomonas* spp. were consistently reported as highly effective for immobilizing cadmium and lead, while *Enterobacter* and *Burkholderia* showed strong potential for copper immobilization. However, their performance remains dependent on the environmental conditions.

Microorganisms such as *Bacillus*, *Pseudomonas*, and *Enterobacter* exhibit high immobilization efficiencies through biosorption, bioprecipitation, bioaccumulation, and biomineralization processes. The effectiveness of these microorganisms is influenced by the environmental conditions, contamination levels, and use of microbial consortia or amendments.

These findings highlight the need for site-specific optimization of microbial treatments to achieve sustainable and effective heavy metal immobilization. Immobilization through microbial pathways—particularly when enhanced with organic amendments—offers a stable, eco-friendly method for reducing metal mobility. However, successful application depends on strain selection, soil compatibility, and environmental adaptation, which highlights the importance of localized optimization.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/land14081547/s1>, Table S1: Efficiency of microorganisms in cadmium immobilization in contaminated soil [27–57,65–67]; Table S2: Efficiency of microorganisms in lead immobilization in contaminated soil [6,29,30,33,40,44–48,51,53,55,56,58–61,66,68–72]; Table S3: Efficiency of microorganisms in copper immobilization in contaminated soil [33,43,44,49,54–57,61–64,72–75].

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Abbreviations

The following abbreviations are used in this manuscript:

MICP	Microbially induced carbonate precipitation
Spp.	Species (plural)
Sp.	Species (singular)

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